

RUTUJA DESHMUKH

AI Engineer | Generative AI | RAG Pipelines | Multi-Agent Systems

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Nashik, 422003

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building production-grade Generative AI applications, Retrieval-Augmented Generation (RAG) pipelines, conversational AI systems, and multi-agent workflows. Proficient in Python, FastAPI, LangChain, LangGraph, ChromaDB, and Streamlit. Skilled in designing scalable AI systems involving semantic retrieval, vector embeddings, workflow orchestration, and intelligent automation. Published researcher (IJARCCE) and national-level presenter with a demonstrated record of delivering end-to-end AI projects from concept to deployment.

TECHNICAL SKILLS

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- **Languages:** Python, SQL, JavaScript
 - **AI / LLM:** RAG, LangChain, LangGraph, Prompt Engineering, Semantic Search, Conversational AI, Vector Embeddings, NLP
 - **ML Frameworks:** scikit-learn, Hugging Face Transformers, Sentence Transformers
 - **Backend:** FastAPI, Django, REST APIs
 - **Frontend:** React.js, Next.js, Streamlit, Tailwind CSS
 - **Databases:** PostgreSQL, SQLite, ChromaDB
 - **Cloud / Tools:** AWS (Cloud Foundations), Git, GitHub, Groq API, Apache Spark
 - **Core Concepts:** Data Structures & Algorithms, ETL Pipelines, Web Crawling, Document Parsing, IoT Integration

EXPERIENCE

Python & Data Engineering Intern | ProWorld Technology Pvt. Ltd. *Jun 2025 – Aug 2025 · Nashik*

- Built Python-based ETL workflows integrated with Django backends, automating data ingestion across 2 live projects — e-commerce and IoT energy monitoring — replacing previously manual export processes.
- Developed web scraping and document parsing modules in Python that processed structured data from external sources into Django-connected databases, reducing the need for manual data entry by the team.
- Integrated REST API endpoints with Django dashboards to enable real-time data retrieval from IoT sensors, making live monitoring available directly inside the analytics interface for the first time.

- Built and delivered 5 responsive UI components across both projects, collaborating with 2 other developers and resolving integration issues identified during QA walkthroughs.
- Documented module behaviour and integration steps for 3 features, helping onboard a new team member mid-internship without requiring senior developer support for those areas.

PROJECTS

AI Multi-Agent Research & Engineering Workspace — *Next.js · FastAPI · LangGraph · LangChain · PostgreSQL · Groq API*

- Built a SaaS-style multi-agent workspace supporting 3 distinct agent roles — research, planning, and coding — each with independent contextual memory and intelligent task routing via LangGraph.
- Designed a FastAPI backend with PostgreSQL to persist agent conversation state across sessions, enabling multi-turn workflows without re-prompting the model for prior context.
- Architected the inter-agent communication layer so that outputs from the research agent are automatically passed as structured context to the planning and coding agents, reducing redundant prompts.

AI Learning Companion — Study Assistant — *Python · Streamlit · ChromaDB · Groq API · Sentence Transformers · RAG*

- Built a fully deployed AI study platform offering 4 core features — PDF chat, AI-generated summaries, adaptive quizzes, and personalised learning roadmaps — accessible via a live Streamlit app.
- Implemented a Hybrid RAG pipeline combining dense vector search (ChromaDB + Sentence Transformers) with keyword retrieval, enabling the system to answer questions from uploaded PDFs with source-grounded responses.
- Integrated a voice interaction module and an analytics dashboard that tracks quiz scores and study progress across sessions, making the platform usable as a complete independent study tool.

Link: [Live Demo](#)

Link: [GitHub](#)

AgroBot — AI-Powered Smart Autonomous Mower — *Python · Raspberry Pi · Node.js · IoT · Android Integration*

- Developed an autonomous mower system controlled by Raspberry Pi, integrating real-time sensor input, motor control logic, and an Android app for remote command dispatch over a local network.
- Presented the complete working prototype at IET Expo 2026 — a national-level technical competition — and published the underlying research in IJARCCCE, an international peer-reviewed journal.
- Implemented the obstacle detection and path-correction logic in Python, allowing the mower to respond to sensor events and adjust movement without manual intervention during operation.

Link: [Research Paper — IJARCCCE](#)

AI-Based Energy Monitoring System — *Django · Python · IoT · Analytics Dashboard*

- Contributed to a Django-based platform that ingested live energy readings from IoT sensors into a structured database and displayed consumption trends on an analytics dashboard.
- Built the data processing pipeline that cleaned and formatted raw sensor output before storage, ensuring the dashboard always displayed consistent, correctly-typed values.

EcoCart — E-Commerce Web Application — *Django · Python · HTML · CSS · JavaScript*

- Developed a fully functional Django e-commerce application covering product catalogue, cart management, and order submission — with all backend logic handled through REST API calls.
- Built responsive frontend templates using HTML, CSS, and JavaScript that adapted cleanly across mobile and desktop screen sizes without external CSS frameworks.

EDUCATION

Diploma — Artificial Intelligence & Machine Learning | **K. K. Wagh Polytechnic** *Jul 2026 (Expected) · Nashik, MH*

Current Score: 91.65% (5th Semester)

Secondary School Certificate (SSC) | **Shree Ram Vidhyalaya** *May 2023 · Nashik, MH*

Score: 92.80%

CERTIFICATIONS

- **Introduction to Generative AI** — *Google Cloud & Simplilearn*
- **Build AI Agents using LangGraph** — *Simplilearn*
- **AWS Academy Graduate — Cloud Foundations** — *Amazon Web Services*
- **Big Data Analytics Projects with Apache Spark** — *Infosys Springboard*
- **Python Training** — *Spoken Tutorial, IIT Bombay*
- **Database Management System** — *Infosys Springboard*

RESEARCH PUBLICATION

AgroBot — AI-Powered Autonomous Grass Mower

International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE) · [View Published Paper](#)

ACHIEVEMENTS & HIGHLIGHTS

- Presented AgroBot at IET Expo 2026, a national-level technical innovation exhibition, competing against final-year B.Tech teams as a diploma student..
- Designed, built, and deployed 5 end-to-end AI-powered applications spanning RAG pipelines, conversational AI, multi-agent orchestration, and IoT-integrated systems.
- Achieved 91.65% in the 5th Semester of Diploma in Artificial Intelligence & Machine Learning.
- Secured 92.80% in the Secondary School Certificate (SSC) examination.